

ETCCCH104 — ENGINEERING CHEMISTRY

UNIT 1 — WATER TECHNOLOGY

Topics: Water analysis; hardness and its determination by EDTA method; internal treatments (phosphate, colloidal, calgon); external treatments (ion-exchange, lime-soda, zeolite); boiler scales — formation, ill effects, prevention; numerical problems.

Section A — Short Answer Questions (2 marks)

1. Define hardness of water and discuss its types.
2. How is the zeolite method different from the ion-exchange process?
3. Differentiate between temporary and permanent hardness with one example each.
4. Define ppm, mg/L, °Fr (French degree) and °Clarke as units of hardness. Give the conversion factors.
5. What is meant by equivalent weight of CaCO_3 ? Why is hardness expressed in terms of CaCO_3 equivalent?
6. What is the principle of phosphate conditioning?
7. What is calgon conditioning? Give the chemical formula of calgon.
8. What is colloidal conditioning? Name two colloidal agents used.
9. What is the role of the buffer in EDTA titration of hardness? Mention its pH.
10. List any two advantages of zeolite over the lime-soda process.
11. State two limitations of the zeolite process.
12. Differentiate between cation-exchange and anion-exchange resins with one example of each.

Section B — Long Answer Questions

1. Explain the EDTA method for determination of hardness of water in detail
2. Discuss the formation of boiler scales and sludge in detail. Explain their ill effects and methods of prevention.
3. Describe the zeolite process for water softening. Include the construction of the unit, softening and regeneration reactions, advantages and disadvantages.
4. Describe the ion-exchange process for water treatment with neat schematic diagrams of the cation- and anion-exchange columns. Include exhaustion and regeneration reactions.
5. Explain the lime-soda process for water softening. Differentiate between cold and hot lime-soda processes with reactions.
6. What are the various internal treatment methods for water softening? Explain any four with suitable reactions.
7. Discuss boiler scale formation and its ill effects.

Section C — Numerical Problems

1. A water sample contains the following dissolved salts per litre: $\text{Ca}(\text{HCO}_3)_2 = 122$ mg, $\text{Mg}(\text{HCO}_3)_2 = 73$ mg, $\text{CaCl}_2 = 55.5$ mg, $\text{MgSO}_4 = 60$ mg. Calculate the temporary hardness, permanent hardness and total hardness of the sample in ppm and °Fr (as CaCO_3). [Mol. wt.: $\text{Ca}(\text{HCO}_3)_2 = 162$, $\text{Mg}(\text{HCO}_3)_2 = 146$, $\text{CaCl}_2 = 111$, $\text{MgSO}_4 = 120$]
2. A water sample contains: $\text{Ca}(\text{HCO}_3)_2 = 162$ mg/L, $\text{Mg}(\text{HCO}_3)_2 = 146$ mg/L, $\text{CaSO}_4 = 136$ mg/L. Determine the (a) temporary, (b) permanent, and (c) total hardness in mg/L and °Clarke. [Mol. wt.: $\text{Ca}(\text{HCO}_3)_2 = 162$, $\text{Mg}(\text{HCO}_3)_2 = 146$, $\text{CaSO}_4 = 136$, $\text{CaCO}_3 = 100$]

3. A water sample contains: $\text{Mg}(\text{HCO}_3)_2 = 14.6 \text{ mg/L}$, $\text{Ca}(\text{HCO}_3)_2 = 16.2 \text{ mg/L}$, $\text{MgSO}_4 = 12.0 \text{ mg/L}$, $\text{CaCl}_2 = 11.1 \text{ mg/L}$, $\text{MgCl}_2 = 9.5 \text{ mg/L}$, $\text{NaCl} = 5.0 \text{ mg/L}$. Calculate the temporary, permanent and total hardness of the water in ppm.
4. A sample of hard water contains 80 mg of CaSO_4 and 40 mg of MgCl_2 dissolved in 2 litres of water. Calculate the total hardness in ppm, °Fr and °Clarke. [Mol. wt.: $\text{CaSO}_4 = 136$, $\text{MgCl}_2 = 95$]
5. 20 mL of a standard hard-water sample (containing 1 mg of CaCO_3 per mL) required 25 mL of EDTA solution. 100 mL of a sample water consumed 18 mL of the same EDTA. After boiling and filtering, 100 mL of the same sample required 10 mL of EDTA. Calculate the total, permanent and temporary hardness of the sample.
6. A zeolite softener (capacity 1000 °Fr·L of CaCO_3 equivalent) is used to soften water having a hardness of 250 ppm. How many litres of water can be softened before regeneration is required?
7. Calculate the amount of lime (90% pure) and soda (98% pure) required to soften 1,00,000 L of water containing: $\text{Ca}(\text{HCO}_3)_2 = 16.2 \text{ mg/L}$, $\text{Mg}(\text{HCO}_3)_2 = 14.6 \text{ mg/L}$, $\text{MgCl}_2 = 9.5 \text{ mg/L}$, $\text{CaSO}_4 = 13.6 \text{ mg/L}$, $\text{MgSO}_4 = 12.0 \text{ mg/L}$.

UNIT 2 — CHEMICAL FUELS

Topics: Classification of fuels; calorific value (HCV & LCV); bomb calorimeter; coal analysis (proximate and ultimate); refining of petroleum; cracking; knocking; anti-knocking agents; LPG, CNG; numerical problems.

Section A — Short Answer Questions (2 marks)

1. What do you understand by calorific value? Explain HCV and LCV.
2. Explain anti-knocking agents with examples.
3. Discuss any two characteristics of an ideal fuel.
4. What do you mean by refining of petroleum?
5. What is the difference between CNG and LPG?
6. What is leaded and unleaded petrol? State the role of TEL.
7. Define proximate analysis of coal. List its components.
8. Define ultimate analysis of coal. List its components.
9. What is meant by knocking in IC engines? State two ill effects.
10. Differentiate between thermal cracking and catalytic cracking.
11. What is the significance of moisture content in coal?

Section B — Long Answer Questions (4 / 8 marks)

1. Discuss the classification of fuels (solid, liquid, gaseous) with suitable examples and advantages.
2. Explain in detail the bomb calorimeter method for determination of calorific value of a fuel. Include the formula used and necessary corrections.
3. How do you determine the nitrogen, sulphur, ash and oxygen content of a coal sample? Explain.
4. What are the advantages of using hydrogen and LPG as a fuel?
5. Explain proximate analysis of coal. Discuss the determination, significance and limitations of each parameter.
6. Explain ultimate analysis of coal. Discuss the determination of carbon, hydrogen, nitrogen, sulphur, oxygen and ash with reactions.
7. Discuss the refining of petroleum and fractional distillation. List the major fractions with their boiling ranges and uses.
8. Explain the mechanism of knocking. Discuss its ill effects, prevention and the role of anti-knocking agents.

Section C — Numerical Problems

1. A fuel has the following composition by mass: C = 80%, H = 10%, O = 5%, S = 2%. Calculate the Higher Calorific Value (HCV) using Dulong's formula. Also calculate the Lower Calorific Value (LCV).
2. In a bomb calorimeter experiment, a solid fuel of mass 1.90 g is burnt; the temperature of 4000 g of water rises by 4.2 °C. The fuel contains 2% H. Calculate the HCV and LCV of the fuel. [Water equivalent of the calorimeter = 385 g; latent heat of steam = 587 cal/g]
3. A coal sample contains: C = 85%, H = 8%, O = 3%, S = 1%, N = 1%, ash = 2%. Calculate the HCV and LCV using Dulong's formula. [Latent heat of steam = 587 cal/g]
4. A sample of coal contains 90% C, 3.5% H, 2.5% O and 4% ash. When this coal was tested in a bomb calorimeter, the following data were obtained: mass of coal = 0.75 g, mass of water = 1500 g, water equivalent of calorimeter = 250 g, rise in temperature = 2.8 °C, cooling correction = +0.05 °C, fuse-wire correction = 8 cal, acid correction = 25 cal. Calculate the HCV and LCV of the coal.

5. 1.5 g of a coal sample on burning in a bomb calorimeter raised the temperature of 1500 g of water from 25.0 °C to 27.5 °C. The water equivalent of the calorimeter is 350 g. The cooling correction is +0.04 °C. If the coal contains 6% H, calculate (a) HCV and (b) LCV. [Latent heat of steam = 587 cal/g]

UNIT 3 — BATTERY TECHNOLOGY

Topics: Galvanic cell; electrode potential; EMF and cell representation; classification of batteries; battery characteristics; Ni-Cd and Lithium-ion batteries; working principle of an electric vehicle.

Section A — Short Answer Questions (2 marks)

1. Define EMF and electrode potential.
2. Define EMF of a cell.
3. List any two examples of a reserve battery.
4. Define a galvanic cell. Give one example with cell representation.
5. Differentiate between primary and secondary batteries with one example of each.
6. Define battery capacity and energy density.
7. Define power density and energy efficiency of a battery.
8. What is meant by the cycle life of a battery?
9. What is shelf life of a battery? On what factors does it depend?
10. Mention any two applications of a Ni-Cd battery.
11. Mention any two applications of a Li-ion battery.
12. State the Nernst equation. Define each term.

Section B — Long Answer Questions (4 / 8 marks)

1. Discuss the constructional features, working and applications of Lithium-ion batteries with a well-labelled diagram.
2. Discuss primary, secondary and reserve batteries with at least two examples of each.
3. Discuss the following characteristics of batteries: voltage, energy density, power density, cycle life, shelf life.
4. Explain the construction, working and reactions of a Ni-Cd battery. State its advantages and limitations.
5. Describe the construction, working and reactions of a lead-acid storage battery (charging and discharging).

Section C — Numerical Problems

1. By using standard electrode potential values, determine whether a reaction is feasible between the following species (attempt any two): (a) $\text{Fe}^{3+}(\text{aq})$ and $\text{Cl}^{-}(\text{aq})$; (b) $\text{Fe}^{3+}(\text{aq})$ and $\text{Br}^{-}(\text{aq})$; (c) $\text{Br}_2(\text{aq})$ and $\text{Fe}^{2+}(\text{aq})$. Given: $2\text{Fe}^{3+} + 2\text{e}^{-} \rightarrow 2\text{Fe}^{2+}$ ($E^{\circ} = +0.77 \text{ V}$); $2\text{Cl}^{-} \rightarrow \text{Cl}_2 + 2\text{e}^{-}$ ($E^{\circ} = +1.36 \text{ V}$); $2\text{Br}^{-} \rightarrow \text{Br}_2 + 2\text{e}^{-}$ ($E^{\circ} = -1.09 \text{ V}$).
2. Calculate the EMF of the cell $\text{Zn} \mid \text{Zn}^{2+} (0.1 \text{ M}) \parallel \text{Cu}^{2+} (1.0 \text{ M}) \mid \text{Cu}$ at 25°C . [$E^{\circ}(\text{Zn}^{2+}/\text{Zn}) = -0.76 \text{ V}$, $E^{\circ}(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$]
3. Calculate the EMF of the cell $\text{Cd} \mid \text{Cd}^{2+} (0.1 \text{ M}) \parallel \text{Ag}^{+} (0.01 \text{ M}) \mid \text{Ag}$ at 25°C . [$E^{\circ}(\text{Cd}^{2+}/\text{Cd}) = -0.40 \text{ V}$, $E^{\circ}(\text{Ag}^{+}/\text{Ag}) = +0.80 \text{ V}$]

UNIT 4 — POLYMERS

Topics: Basic concepts of polymers; types; thermoplastics & thermosetting plastics; preparation and applications of industrially important polymers (Natural rubber, Buna-S, Buna-N, Neoprene, Isoprene, Nylon-6, Nylon-6,6, Dacron, Terylene); conducting and biodegradable polymers.

Section A — Short Answer Questions (2 marks)

1. What are thermoplastics? Give examples.
2. What is polymerization? Mention its types.
3. How are polymers classified according to molecular forces?
4. What do you mean by vulcanization of rubbers?
5. Differentiate between addition and condensation polymerization with one example of each.
6. What are conducting polymers? Give two examples.
7. What are biodegradable polymers? Give two examples.
8. Write the monomers and the repeat unit of Neoprene.
9. Mention any two industrial applications of natural rubber.

Section B — Long Answer Questions (4 / 8 marks)

1. Differentiate between thermoplastic and thermosetting polymers (at least four points each).
2. Discuss the preparation and uses of Buna-S and Neoprene rubber.
3. Discuss the preparation and applications of Buna-S and Buna-N.
4. Explain the preparation, properties and applications of Nylon-6,6.
5. Explain the preparation, properties and applications of biodegradable polymers along with suitable examples.
6. Classify polymers on the basis of: (a) source, (b) structure, (c) molecular forces, (d) mode of polymerization. Give examples in each case.